Phase 2: Architecture, Prototype, and Evaluation Plan

Multi-Agent Genealogical Research System with Adversarial Critique and DNA Verification

Field	Details
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Track	Track A: Technical Build
Course	94815 Agentic Technologies
Semester	Spring 2026

Architecture Overview

The system is currently a four-agent sequential pipeline with an adversarial feedback loop, orchestrated by LangGraph. Each agent performs a distinct function in the genealogical research workflow:

- retrieving evidence
- consolidating evidence into profiles
- proposing relationships
- independently critiquing proposed relationships

A fifth agent (DNA Analyst) is scoped for Phase 3. The architecture's central design claim is that a single LLM cannot meaningfully critique its own genealogical conclusions due to confirmation bias. Structural separation between hypothesis generation and adversarial critique, with an enforced information isolation boundary, provides genuine independent review. This mirrors both AI safety red-teaming methodology and the professional genealogical peer review process defined by the Genealogical Proof Standard.

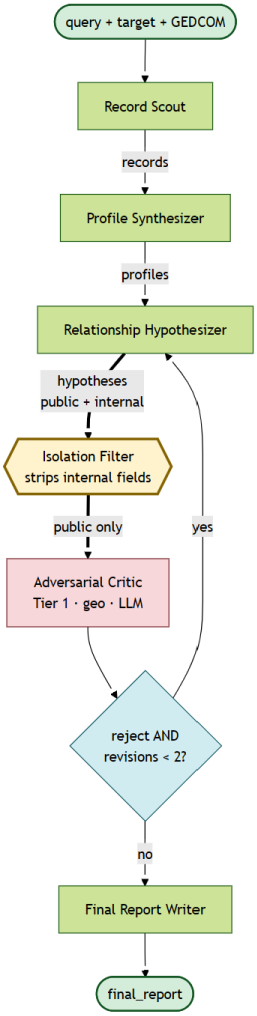

Figure 1. This diagram shows the current Phase 2 system architecture. The isolation boundary (yellow) enforces a whitelist filter between the Hypothesizer and Critic, stripping internal reasoning fields to prevent confirmation bias. The conditional router permits up to two revision cycles before forced finalization. All agents share state via LangGraph; edge labels indicate the primary state fields flowing between nodes.

Role Definitions

Record Scout

Purpose
Retrieve candidate historical records for a target person. Retrieves but does not interpret. All downstream analysis is performed by other agents.

Inputs

- query (natural language research question)
- target_person (name, approximate dates, locations)
- gedcom_text (raw GEDCOM file content)

Process
The LLM extracts structured search criteria from the query, including alternate name forms (e.g., “Philip II of Spain” also mapped to “Philipp II”, “Felipe II”). The Scout then parses the GEDCOM into structured person records using python-gedcom-2, runs fuzzy name matching (Jellyfish: Soundex, Metaphone, Jaro-Winkler, Levenshtein) against all persons, and returns the top candidates with their family context.

Outputs

- gedcom_persons (all parsed persons)
- retrieved_records (ranked candidate records)
    - composite match scores
    - source citations as record_id
    - family context for each candidate

Tools

- tools/gedcom_parser.py (GEDCOM to structured dicts)
- tools/fuzzy_match.py (composite name similarity scoring)
- Claude Sonnet (search criteria extraction and alt-name generation)

Design Note
The Scout deliberately surfaces multiple fuzzy matches above threshold rather than selecting one. Disambiguation is the Synthesizer’s responsibility. In the Kennedy test, this surfaced a 19th-century “John Kennedy” infant alongside JFK, which is exactly the kind of candidate that would cause a false merge if resolved prematurely.

Profile Synthesizer

Purpose
Consolidate retrieved records into a single sourced person profile with explicit citations and gap flagging. Resolve which candidate record actually matches the query target.

Inputs

- retrieved_records
- target_person

- query

Process
Disambiguation is deterministic and pre-LLM. Candidates are scored by name similarity, birth year proximity, and location token overlap. The highest-scoring candidate is selected; all others are explicitly excluded with scored reasons. The LLM then consolidates facts from the selected record into a structured profile, with every fact citing its source record_id. Gaps are flagged explicitly.

Outputs

- profiles (list containing the consolidated profile)
    - disambiguation: candidates_considered with scores and exclusion reasons for each
    - facts: cited biographical facts, each with a source record_id
    - family_references: father, mother, spouse, children, each with name and record_id
    - gaps: explicitly flagged missing information (e.g., no marriage date, single-source corroboration)

Tools

- tools/fuzzy_match.py (disambiguation scoring)
- tools/date_utils.py (date normalization)
- Claude Sonnet (fact consolidation and gap identification)

Design Note
The disambiguation.candidates_considered block is evidence that the Critic can inspect. It records every candidate's score and exclusion reason, making the identity resolution decision auditable. This is a deliberate departure from opaque LLM-based disambiguation because the math is reproducible.

Relationship Hypothesizer

Purpose
Propose family connections with evidence chains, confidence scores, and stated weaknesses. Must acknowledge the limitations of its own hypotheses.

Inputs

- profiles
- gedcom_persons
- retrieved_records

Process
Routes query to relationship types (parents, spouses, children, siblings) via keyword matching. For each candidate relationship, computes deterministic prior evidence: birth-year-derived age delta, haversine birthplace distance, biological plausibility flags. Then calls the LLM once per relationship with the subject's profile, candidate person data, computed priors, and explicit allowed citation sources. Uses the make_hypothesis() constructor from

agents/hypothesis_schema.py, which enforces population of both public and internal field sets

Outputs

- hypotheses (list of hypothesis dicts)
    - public fields (visible to Critic)
        - hypothesis_id, subject_id, related_id, proposed_relationship, evidence_chain, confidence_score, stated_weaknesses.
    - internal fields (hidden from Critic)
        - reasoning_narrative, alternatives_considered, intermediate_steps, llm_raw_response

Tools

- tools/date_utils.py (age-at-event calculation)
- tools/geo_utils.py (haversine distance)
- Claude Sonnet (evidence synthesis and confidence assessment)

Design Note

The Hypothesizer's make_hypothesis() constructor enforces population of both public and internal field sets at creation time, so it's impossible to accidentally create a hypothesis without the fields the isolation filter expects.

Adversarial Critic

Purpose

Independently evaluate hypotheses without access to the Hypothesizer's reasoning. Catch errors, flag insufficient evidence, and prevent false merges.

Inputs

- Filtered hypotheses (public fields only
    - via filter_hypothesis_for_critic())
- retrieved_records
- profiles
- gedcom_persons

Process

The Critic executes a three-stage evaluation in fixed order:

Stage 1 | Tier 1 deterministic checks (fail-fast). Runs run_all_tier1_checks() from tools/date_utils.py: death before birth, parent younger than child, implausible lifespans, marriage under age, parent deceased before conception. If any check returns verdict: impossible, the Critic rejects immediately without invoking the LLM. Confidence: 0.98.

Stage 2 | Geographic plausibility (soft flag). Runs check_geographic_plausibility() from tools/geo_utils.py with multi-tier thresholds: <500km ok, 500–1500km ok, 1500–3000km flag_moderate, >3000km flag_strong. Raw distance in kilometers is always included so the LLM can reason about era-appropriate migration patterns.

Stage 3 | LLM evidence sufficiency. The LLM evaluates whether the evidence chain supports the proposed relationship, considering source quality, corroboration, and gaps. It produces a verdict (accept, reject, flag_uncertain), issues found, and confidence_in_critique (0–1).

Outputs

- critiques (list of critique dicts with verdict)
- issues_found
- confidence_in_critique
- tier1_results
- geo_check
- isolation_mode)
- Also increments revision_count and sets status to needs_revision on reject

Tools

- tools/date_utils.py (Tier 1 checks via run_all_tier1_checks())
- tools/geo_utils.py (multi-tier geographic plausibility)
- agents/hypothesis_schema.py (filter_hypothesis_for_critic())
- Claude Sonnet (evidence sufficiency reasoning)

Design Note

Deterministic checks run before the LLM, not alongside it. If Tier 1 finds an impossibility, the LLM is never invoked. This supports fail-fast design. The Critic produces confidence_in_critique on its own output, which is the measurable axis for Tier 3 overconfidence detection. Geographic verdicts are soft flags, never auto-rejects. Era-dependent interpretation is delegated to the LLM rather than encoded in thresholds.

Final Report Writer

Purpose

Note: This is a deterministic step and not an agent. Its purpose is to compose a structured markdown report from the pipeline output. It aggregates profiles, hypotheses, and critiques into a human-readable document.

Coordination Logic

Sequencing

The pipeline is strictly sequential:

Record Scout → Profile Synthesizer → Relationship Hypothesizer → Adversarial Critic → (conditional loop or Final Report Writer)

Each agent reads from shared LangGraph state and writes its outputs back to state. No agent executes until its predecessor has completed.

Adversarial Loop

After the Critic runs, a conditional router evaluates two conditions:

1. Did any hypothesis receive a reject verdict?
2. Is revision_count < 2?

If both are true, the state is routed back to the Hypothesizer with status=needs_revision and revision_count incremented. The Hypothesizer receives the critique and attempts to revise its hypothesis with additional evidence or adjusted confidence.

After 2 rejection cycles, the pipeline forces finalization regardless of verdict. Unsupported hypotheses are marked as such in the final report rather than dropped to preserve the evidence trail for human review.

Isolation Boundary

Between the Hypothesizer and the Critic, filter_hypothesis_for_critic() strips all internal fields from hypothesis dicts. This is a whitelist filter (only explicitly listed public fields pass through), not a blacklist, so any new fields added to hypotheses are hidden by default. The filter is implemented in agents/hypothesis_schema.py and is the single toggle point for the Critic isolation A/B experiment.

The isolation design was validated empirically. See the Evaluation Plan for A/B experiment results.

Stopping Conditions

The pipeline terminates when either:

1. The Critic accepts all hypotheses (status=complete).
2. The revision count reaches 2 (forced finalization).
3. No hypotheses were generated (empty input).

There is no infinite loop risk since the revision counter is a hard ceiling.

Human Intervention Points

The current prototype does not include human-in-the-loop checkpoints. In a production system, natural intervention points would be:

- after disambiguation (user confirms the correct person was selected)
- after hypothesis generation (user provides additional context)
- after the final report (user accepts or requests further research)

These are documented as Phase 3 scope.

Tools, Memory, and Data Design

Tools

Tool	Used by	Purpose
gedcom_parser.py	Record Scout	Parse GEDCOM files into structured person dicts
fuzzy_match.py	Scout, Synthesizer	Name similarity scoring (Soundex, Metaphone, Jaro-Winkler, Levenshtein)
date_utils.py	Critic, Hypothesizer	Date normalization with uncertainty envelopes; Tier 1 impossibility checks
geo_utils.py	Critic, Hypothesizer	Haversine distance, geocoding (Nominatim), multi-tier geographic plausibility
trace_writer.py	main.py	Persist interaction traces to JSON + markdown files

State Schema

State is managed as a LangGraph TypedDict passed between all nodes:

Field	Type	Written by	Read by
query	str	Entry point	Scout, Synthesizer, Hypothesizer
target_person	dict	Scout (LLM)	Synthesizer, Hypothesizer
gedcom_persons	list[dict]	Scout (parser)	All downstream agents
retrieved_records	list[dict]	Scout	Synth, Hypo, Critic
profiles	list[dict]	Synthesizer	Hypothesizer, Critic
hypotheses	list[dict]	Hypothesizer	Critic (filtered view)
critiques	list[dict]	Critic	Report Writer
revision_count	int	Critic	Router
status	str	Critic	Router
final_report	str	Report Writer	End

trace_log	list[str]	All agents	Trace writer

Data Sources

Primary

GEDCOM file exports (Ancestry, public sample files). The parser handles GEDCOM 5.5 format with strict parsing disabled to accommodate formatting variations across genealogy software.

Tested Against

- Kennedy Family (70 persons, English, UTF-8)
- Moore Family Tree (8,759 persons, English, UTF-8, developer's own tree)
- Queen Victoria lineage (4,683 persons, English, UTF-8)
- Habsburg dynasty (34,020 persons, German, Latin-1 encoding).

Phase 3

- DNA match exports with shared cM values (.csv)
- web search targeting FamilySearch.org, FindAGrave, WikiTree

Memory Design

The current system uses session-only state, so there is no cross-run persistence. All information flows through the LangGraph state object within a single pipeline invocation. The trace_log field provides an append-only audit trail, and completed traces are persisted to traces/ as JSON and markdown files for post-run analysis.

This is a deliberate simplicity choice. Persistent memory (e.g., accumulating research context across multiple queries about the same family) would be a consideration for future work, but is out of scope for this project. It adds architectural complexity without improving the core hypothesis-critique loop being evaluated in this phase.

Why This Architecture, Not a Simpler One

A single-agent system could perform all of these functions in one prompt chain. The multi-agent design earns its place for three reasons:

Confirmation bias prevention. A single LLM that generates a hypothesis and then critiques it in the same context window is structurally incapable of independent review. The isolation boundary between the Hypothesizer and Critic is not just a design preference—it is the mechanism by which the system achieves genuine adversarial evaluation. The A/B experiment demonstrates that reasoning visibility measurably increases linguistic leakage from the Hypothesizer's narrative into the Critic's output.

Separation of deterministic and probabilistic reasoning. The Critic's Tier 1 checks are deterministic rule-based validators that must execute before LLM reasoning, not alongside it. If a parent's birth date is after their child's, no amount of LLM reasoning should override

that. The fail-fast architecture guarantees deterministic checks run first, and the Tier 1 test cases confirm the LLM is never invoked when a deterministic impossibility is detected.

Auditable evidence chains. Each agent produces structured, inspectable output. The Synthesizer's disambiguation block records why each candidate was selected or excluded. The Hypothesizer's evidence chains cite specific source records. The Critic's issues list references specific evidence gaps. A monolithic system could produce similar output, but the structural separation ensures each stage's reasoning is independently verifiable, The Critic cannot retroactively adjust the Synthesizer's disambiguation to support its preferred conclusion.

Changes from Phase 1

There were two changes from the Phase 1 architecture driven by Phase 1 feedback:

Adversarial trap difficulty calibration. Phase 1 specified adversarial traps but did not calibrate difficulty. Phase 2 implements a three-tier taxonomy (deterministic / plausible-but-wrong / genuinely ambiguous) with tier-specific success criteria. See the Evaluation Plan for details.

Critic isolation empirical validation. Phase 1 proposed withholding the Hypothesizer's reasoning chain from the Critic as a design decision. Phase 2 validates this empirically through an A/B experiment comparing filtered vs. unfiltered Critic behavior. See the Evaluation Plan for results.

I added one additional structural element not in Phase 1:

Geographic plausibility check. This was redesigned from a single binary threshold (3000km) to a multi-tier soft-flag system (<500km ok, 500–1500km ok, 1500–3000km flag_moderate, >3000km flag_strong) after I observed high flag output in sample testing. While the scope of the project is focused in the United States, I am using European lineages, whose generational migrations manifest at a different scale. The era-sensitive interpretation is delegated to the LLM rather than encoded in thresholds.

Prototype Evidence

The core four-agent pipeline runs end-to-end on all tested datasets:

Dataset	Persons	Language	Correct?	Critic verdict	Notable
Kennedy	70	English	Yes	flag_uncertain (0.72)	Single-source caution
Moore	8,759	English	Yes	flag_uncertain (0.72)	Irish migration detected
Queen	4,683	English	Yes	accept (0.72–0.78)	Low name sim, rescued by year
Habsburg	34,020	German	Yes	accept (0.82–0.85)	Cross-language via LLM

Evaluation Plan

Evaluation Strategy

The evaluation tests two claims:

(1) The pipeline produces correct genealogical conclusions from GEDCOM evidence.
(2) The Adversarial Critic catches errors that would otherwise propagate to the user.

Both claims are tested against data with known ground truth, using adversarial trap cases of calibrated difficulty.

A separate architectural validation experiment tests the Critic isolation design (whether withholding the Hypothesizer's reasoning chain from the Critic preserves critique quality while preventing confirmation bias).

Adversarial Trap Taxonomy

Phase 1 feedback identified that the 80% Critic detection target is meaningful only if the traps are realistic and their difficulty distribution is explicit. The evaluation implements a three-tier taxonomy with tier-specific success criteria.

Tier 1 | Deterministic (Rule-Checkable)

Errors detectable by date arithmetic and logical rules alone, with no LLM reasoning required. These validate that the Critic's deterministic checks function correctly and execute before the LLM (fail-fast pattern).

Case	File	Planted Error	Expected Check
T1-1	tier1_parent_born_after_child.ged	Parent birth 30y after child	parent_younger_than_child
T1-2	tier1_death_before_birth.ged	Death date precedes birth	death_before_birth
T1-3	tier1_parent_too_young.ged	Mother age 8 at birth	parent_too_young_at_birth

Success criterion: ~100% detection rate. All Tier 1 traps must trigger a reject verdict via the deterministic fail-fast path. The LLM must not be invoked.

Results: 3/3 pass. All three cases rejected via fail-fast with confidence 0.98. The LLM was confirmed not invoked in any Tier 1 rejection.

Tier 2 | Plausible but Wrong (LLM-Catchable)

Errors that look reasonable at first glance but do not survive evidence cross-referencing. Tier 1 checks deliberately do not fire on these cases.

Case	File	Planted error	Signal
T2-1	tier2_geographically_impossible.ged	Palermo ↔ Dublin, 1850	~2000km, no migration context
T2-2	tier2_surname_mismatch.ged	Patronymic inconsistency	Surname logic violation
T2-3	tier2_implausible_age_gap.ged	85-year father-child gap	Biologically implausible

Success criterion: ≥80% detection rate via LLM reasoning

Results: 3/3 pass. All caught by LLM critique. The implausible age gap case triggered the full revision loop (both Critic runs rejected, graph forced finalize after 2 revisions), correct behavior.

Tier 3 | Genuinely Ambiguous (Overconfidence Detection)

Cases where reasonable genealogists could disagree. The Critic should express appropriate uncertainty rather than issuing a confident accept or reject.

Case	File	Planted ambiguity	Why it's hard
T3-1	tier3_kennedy_duplicate_john.ged	Two "John Kennedy", same era	Plausible either way
T3-2	tier3_kennedy_ambiguous_joseph.ged	Father/son share exact name	Real-world naming pattern

Success criterion (revised): No accept or reject verdict with confidence_in_critique ≥ 0.85 on Tier 3 cases. A flag_uncertain verdict at any confidence level is a pass.

Criterion revision note: *The initial criterion required confidence < 0.70 regardless of verdict. This conflated confidence in the verdict with confidence in the answer. A Critic that says "I'm 0.72 confident this is uncertain" is performing correctly. The revised criterion distinguishes these: high confidence on a flag_uncertain verdict is acceptable; high confidence on an accept or reject of ambiguous evidence is the failure mode.*

Results: 2/2 pass. Both cases produced flag_uncertain verdicts with confidence 0.72.

Known limitation: *Tier 3 passes are partially confounded with the Critic's high base rate of flag_uncertain on single-source GEDCOM files. A cleaner measurement would inject ambiguity into multi-source scenarios where the base acceptance rate is higher. Planned for Phase 3.*

Trap Evaluation Summary

Tier	Cases	Target	Result	What It Validates
1: Deterministic	3	~100%	3/3	Fail-fast rule checks; LLM not invoked
2: Plausible but Wrong	3	≥80%	3/3	Evidence cross-referencing via LLM
3: Ambiguous	2	No overconfident accept/reject	2/2	Calibrated confidence, appropriate uncertainty
Total	**8**		**8/8**	

Reconstruction Tests

Independent of adversarial traps, the pipeline was tested end-to-end on four GEDCOM datasets with known correct answers:

Dataset	Persons	Correct?	Critic Verdict	Notable
Kennedy	70	Yes	flag_uncertain (0.72)	19th-century namesake surfaced; correctly disambiguated
Moore	8,759	Yes	flag_uncertain (0.72)	Trans-Atlantic migration detected; 3-way name collision handled
Queen	4,683	Yes	accept (0.72–0.78)	Low name similarity rescued by birth year; GEDCOM error caught
Habsburg	34,020	Yes	accept (0.82–0.85)	Cross-language matching via LLM alt-names

Observed pattern: Critic confidence correlates with evidence density and name specificity, not tree size. Habsburg and Queen received accept verdicts (denser cross-references, distinctive historical names). Kennedy and Moore received flag_uncertain (single GEDCOM source, common surnames). The system's confidence scales with available evidence.

Critic Isolation A/B Experiment

Design

The Phase 1 architecture proposed withholding the Hypothesizer's reasoning chain from the Critic to prevent confirmation bias. Phase 1 feedback recommended validating this empirically.

Setup: For each test query, the pipeline runs through Record Scout → Profile Synthesizer → Relationship Hypothesizer, producing a shared seed state. The Critic then runs twice against identical hypotheses:

- **Condition A (filtered):** Critic receives hypotheses with internal fields stripped via filter_hypothesis_for_critic(). This is the production design.
- **Condition B (unfiltered):** Critic receives hypotheses with all fields visible, including reasoning_narrative, alternatives_considered, and intermediate_steps.
- **Toggle mechanism:** A single constant in adversarial_critic.py switches between conditions. Every critique records its isolation_mode field, making results self-labeling.

Queries Tested

Three queries spanning difficulty tiers

- JFK | parents (well-documented reconstruction)
- Tier 2 | surname mismatch (plausible-but-wrong trap)
- Tier 3 | ambiguous Joseph Kennedy (genuinely ambiguous)

Results

Query	Hyp	A verdict	A conf	B verdict	B conf	Leak A	Leak B
JFK	father	flag_uncertain	0.72	flag_uncertain	0.72	1	2
JFK	mother	accept	0.72	accept	0.72	0	0
Tier 2	father	flag_uncertain	0.78	flag_uncertain	0.72	1	3
Tier 2	mother	flag_uncertain	0.72	flag_uncertain	0.72	1	3
Tier 3	father	accept	0.72	flag_uncertain	0.72	3	4
Tier 3	mother	flag_uncertain	0.72	flag_uncertain	0.72	1	1

Aggregate leakage: Condition A: 7. Condition B: 13.

Findings

Isolation does not degrade Critic performance. 5/6 hypotheses received identical verdicts across conditions. Confidence scores were within 0.06 of each other. The Critic produces useful critiques without access to the Hypothesizer's reasoning.

Reasoning visibility increases linguistic leakage. Condition B produced nearly twice the 5-gram phrase overlap (13 vs. 7). This is consistent across all three queries and matches the confirmation bias hypothesis.

The one verdict change was surprising. On the Tier 3 case, the Critic in Condition A (isolated) issued accept at 0.72, while Condition B switched to flag_uncertain. This is the opposite of the confirmation bias prediction: the Critic became more skeptical with more information. The Hypothesizer's reasoning_narrative included explicitly stated weaknesses, and the Critic absorbed those doubts rather than being anchored by the conclusion.

Conclusion: The isolation design is validated as safe (no performance degradation) and beneficial (reduced linguistic leakage). The surprising Tier 3 result suggests a potential Phase 3 refinement: passing a structured evidence summary without the full reasoning narrative.

Limitations

- Small sample (6 hypotheses across 3 queries)
- Overdetermined cases (JFK evidence may predetermine verdicts)
- Crude leakage metric (5-gram overlap misses paraphrasing)
- Single LLM (results may not generalize across models).

Success Criteria Summary

Metric	Phase 1 target	Phase 2 result	Status
Record retrieval	≥80%	4/4 correct person in set	On track
Citation accuracy	Zero fabricated	No fabrications observed	On track
Hypothesis accuracy	≥70% top-ranked	4/4 correct parents	On track
Adversarial detection (T1+T2)	≥80%	6/6 detected (100%)	Exceeding
Adversarial detection (T3)	Calibrated confidence	2/2 appropriate uncertainty	On track
False merge rate	≤15%	0 false merges	On track

Phase 3 Evaluation Extensions

Cleaner Tier 3 measurement: Inject ambiguity into multi-source scenarios where the Critic's base acceptance rate is higher, isolating genuine ambiguity detection from single-source caution.

DNA consistency tests: Test cases where shared cM values are inconsistent with proposed documentary relationships.

Ablation comparison: Full pipeline vs. single-agent baseline on the same queries to quantify multi-agent design value.

Expanded A/B experiment: More queries, embedding-based leakage metric, and testing the structured-evidence-summary middle ground.

Developer's family tree evaluation: Moore Family Tree GEDCOM with the my verified family knowledge as ground truth to minimize contamination risk that may exist for more famous lineages.

Risk and Governance Plan

Failure Modes

Observed During Phase 2 Implementation

Non-UTF-8 GEDCOM files crash the parser. The Habsburg.ged file (Latin-1 encoding) caused a UnicodeDecodeError because python-gedcom-2 hardcodes UTF-8 decoding.

> **Mitigation:** A workaround reads the file as Latin-1 and passes it through parse_gedcom_text(), which round-trips through a UTF-8 temp file. A permanent fix would detect encoding from the GEDCOM HEAD CHAR tag.
>
> **Severity:** blocking without workaround, low with workaround.

Royal naming conventions defeat surname-based matching. The Habsburg GEDCOM stores "Philipp II von /Spanien/" (using a geographic title as the surname field). No one in the tree has surname "Habsburg." The Scout compensated through LLM-generated alt-names, but a query for "Habsburg family" would find zero surname matches.

> **Severity:** low for well-known figures, potentially higher for lesser-known individuals.

LLM brings outside world knowledge into the Critic. On the Queen Victoria run, the Critic noted that "the Duchess of Kent had other children from her first marriage." This information is not present in the GEDCOM or any retrieved record. This is historically accurate but violates the principle that the Critic should reason from provided evidence only.

> **Severity:** low in prototype context, but could introduce incorrect claims for less well-documented families.

Anticipated but Not Yet Observed

False merge on common names in large trees. The disambiguation scoring correctly resolved all tested cases, including three-way collisions. However, two individuals with identical names, overlapping birth years, and the same location could defeat the scoring. The Critic's Tier 3 behavior (flagging uncertainty) provides a partial safety net. Severity: medium.

Hallucinated citations. The agents are prompted to cite only from provided record_ids, and citation validation checks are implemented. No hallucinated citations were observed. However, under sparse evidence conditions, the LLM could fabricate a plausible-sounding record_id. Severity: high if undetected. Mitigation: citation validation against the known record_id set with trace-level warnings.

Adversarial loop stalls on borderline cases. If the Hypothesizer produces a hypothesis that the Critic repeatedly rejects but cannot improve, the system exhausts its 2 revision cycles and forces finalization. The hypothesis is marked as unsupported rather than dropped. Severity: low. Mitigation: explicit labeling in the final report.

Geographic check false negatives on migration-era genealogies. The multi-tier system flags distances above 1500km. For immigration-era families, parent-child distances of 4000–5000km are routine. The LLM interprets flags relative to birth year, but missing or

incorrect dates could cause over-weighting. Severity: low (observed to work correctly on the Moore tree's Irish-American case).

Privacy and Data Sensitivity

Family Data

GEDCOM files and DNA match exports contain personally identifiable information: names, birth/death dates, birth/death places, marriage records, and genetic relationship data. For living individuals, this data is sensitive.

Current mitigations: GEDCOM files stored in data/PII Trees/ and excluded from version control. Trace writer replaces raw GEDCOM text with a SHA-256 hash. Parsed person records are retained in traces (names and dates visible). No family data transmitted externally except to the Anthropic API (LLM calls) and Nominatim (geocoding place names only). Developer's family tree used with data owner's knowledge and consent.

Phase 3 consideration: *If deployed beyond the developer's own use, a data consent and retention policy would be required. Users should understand that GEDCOM data is sent to the LLM provider's API and that traces containing parsed person records are stored locally.*

DNA Data

DNA match exports contain shared cM values and predicted relationships. While individual cM values are not directly identifying, the combination of match patterns across a full export can uniquely identify an individual. DNA data will be handled with the same protections as GEDCOM data: local storage, excluded from version control, not included in submission artifacts.

API Key Security

The Anthropic API key is stored in .env, excluded from version control, and loaded via python-dotenv. A .env.example file with a placeholder value is committed for setup instructions. No API keys appear in traces, logs, or submission artifacts.

Trust and Reliability

Confidence Calibration

The system produces confidence scores at two levels: the Hypothesizer's confidence in its proposed relationship, and the Critic's confidence in its own critique. These are LLM-generated and should not be interpreted as calibrated probabilities. Observed behavior suggests reasonable ordinal ranking (higher evidence density produces higher confidence) but absolute values are not statistically calibrated.

User-facing risk: A genealogist unfamiliar with LLM confidence scores might interpret "0.85 confidence" as "85% probability of being correct." The system does not make this claim. The final report should include a note that confidence scores reflect evidence strength assessment, not calibrated probability.

Single-Source Limitation

The Phase 2 prototype operates primarily on GEDCOM data, a single source that may itself contain errors. The Critic correctly identifies this limitation (flagging single-source

uncertainty in every critique), but the system cannot currently corroborate GEDCOM claims against independent sources. Phase 3's web search integration and DNA analysis will partially address this.

LLM Dependence

Four of the five pipeline stages use LLM calls. LLM behavior is non-deterministic, so the same input can produce different confidence scores, different issue lists, and occasionally different verdicts across runs. Observed variance is small (confidence scores within ±0.06 on the same input), but the system is not fully reproducible. The deterministic components (Tier 1 checks, disambiguation scoring, geographic distance) are fully reproducible.

Mitigation: Traces capture the full LLM input and output for every call. The A/B experiment demonstrates that architectural decisions (isolation boundary) have a larger effect on system behavior than run-to-run LLM variance.

Misuse Risks

Genealogical Fraud

A user could use the system to construct a fabricated family lineage by feeding it a deliberately falsified GEDCOM file. The system would process the fabricated data and produce a professional-looking report with citations, lending an appearance of rigor to fraudulent genealogy. The Critic would flag evidence gaps but would not detect that the underlying data was fabricated, since it evaluates internal consistency rather than external validity.

Mitigation: The system's citations always point back to the input GEDCOM. A reviewer can trace any claim to its source record and verify legitimacy independently. The system does not and cannot verify the authenticity of its input data.

Privacy Violations via Genealogical Inference

The system could be used to infer family relationships that individuals have not disclosed. For example, one could identify biological parents in adoption cases, or surface extramarital relationships visible in DNA data but not in documentary records.

Mitigation: The Phase 3 DNA Analyst should include a warning when relationship predictions conflict with documentary records, as such conflicts may involve sensitive family situations. The system should not present inferred relationships as established facts without explicit uncertainty markers.

Weaponized Genealogy

Historical and ongoing harms have been perpetrated using genealogical claims — racial classification, inheritance fraud, identity theft, and discriminatory lineage requirements. An automated system producing authoritative-looking reports could lower the barrier to such misuse.

Mitigation: The final report includes explicit uncertainty markers, evidence gaps, and the Critic's objections. It does not produce certificates or attestations of lineage. The report supports further research, not definitive legal or identity claims.

Governance Controls

Current Controls

Control	Implementation	Purpose
Revision limit	revision_count < 2 in graph router	Prevents infinite loops
Isolation boundary	filter_hypothesis_for_critic() whitelist	Prevents confirmation bias
Tier 1 fail-fast	Deterministic checks before LLM	LLM cannot override impossibilities
Citation validation	record_id check against known set	Prevents hallucinated citations
Trace logging	Append-only trace_log; JSON/MD persist	Full audit trail for every run
Data isolation	.gitignore excludes .env, PII, traces	Prevents accidental exposure

Controls Planned for Phase 3

Control	Purpose
Human confirmation step	User reviews Critic assessment before report generation
DNA sensitivity warning	Alert when DNA-inferred relationships conflict with documents
Confidence disclaimer	Explicit note that scores are not calibrated probabilities
Input validation	Reject oversized GEDCOM files to prevent resource exhaustion